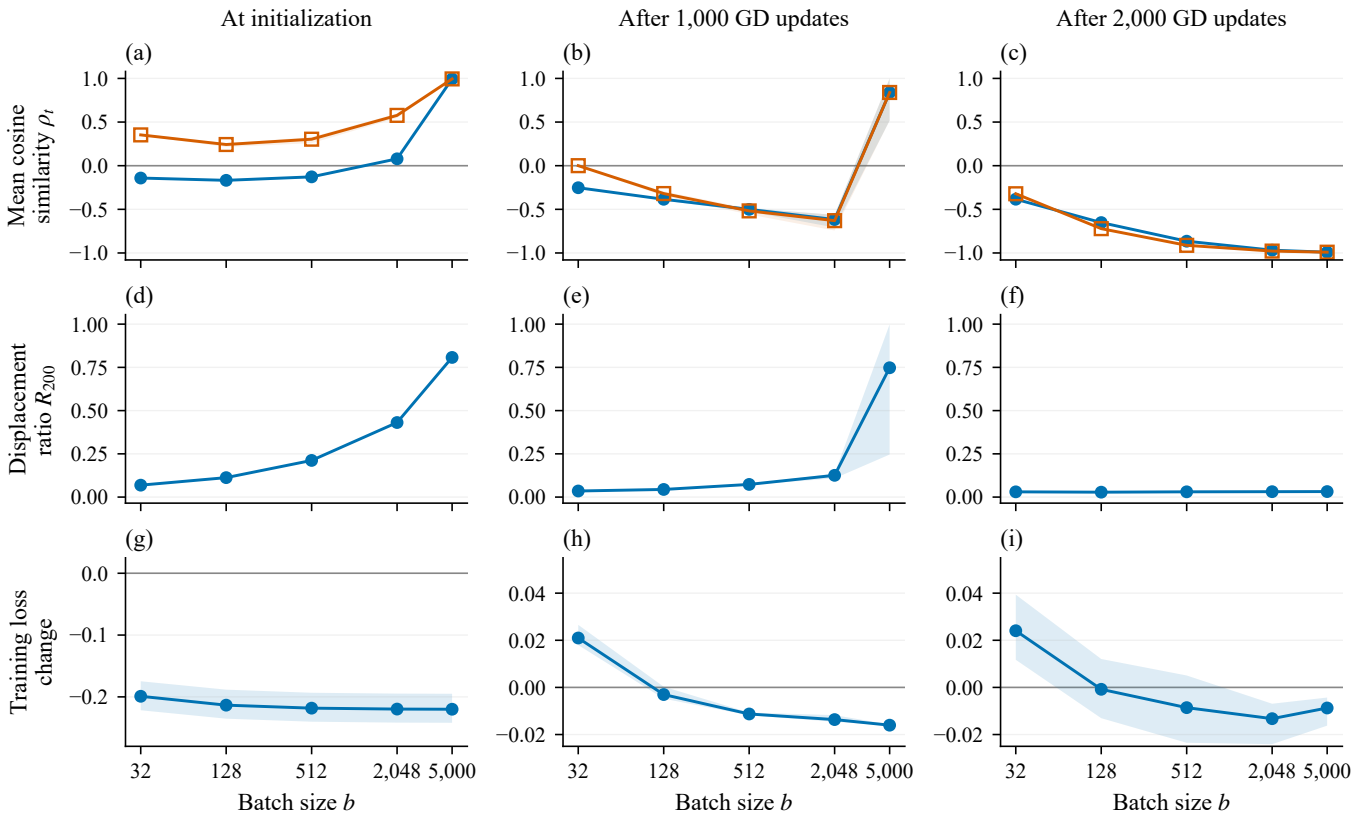

At initialization
After 1,000 GD updates
After 2,000 GD updates
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
Mean cosine similarity ρ_t
Displacement ratio R_{200}
Training loss change
1.0
0.5
0.0
−0.5
−1.0
1.00
0.75
0.50
0.25
0.00
0.04
0.02
0.00
−0.02
−0.1
−0.2
32
128
512
2,048
5,000
Batch size b
Cosine: mini-batch gradients
Cosine: full gradients at the same SGD states